AI for Environmental Degradation and Sustainable Development

I. Abstract

For years average temperature has been progressively rising on our planet, namely from CO2 emissions[5]. This issue has reached a critical juncture where the implications of our actions or lack there of may have reached a point where damage is being done beyond repair. Experts posit that an average global increase beyond 1.5 degrees Celsius beyond that of the pre-industrial period would have detrimental impacts on many different aspects of our modern world but especially on our natural one[4]. In this study we will use data analytics and machine learning techniques to analyze comprehensive datasets of environmental metrics. Polynomial Regression is used to project future temperature trends while use of a heat-mapped correlation matrix explores the complexities between environmental and socio-economic variables. This analysis revealed a variety of outputs suggesting complex relationships, underscoring the importance of future in depth machine learning implementation.

II. Introduction

Technical evolution is a double-edged sword. Each step generates several new pathways that could lead to great bounds forward or even create new problems. One such paradoxical instance is the use of AI in reference to environmental studies. This data monitoring is immensely important to our ability to combat the damages being done to the environment. Machine learning is used to monitor environmental degradation often caused by the same pollution that is integral in the power usage of these models.

While there is not an all-encapsulating definition of AI, we will consider it any software technology with at least 1 of the following functions: Pattern recognition, prediction, decision making or perception such as image analysis [1]. Experts posit that these AI applications will help us achieve 134 sustainable development goals (SDGs) while inhibiting 59 other targeted improvements [2]. Additionally, the same technology can be used to optimize renewable energy solutions that are largely dependent on the same climate data.

Leaders in this field have taken previous studies conducted by various sources and mapped sustainable development goals correlation and proposed improvement via AI [3]. As seen in this data, AI stands to improve 79% of all reviewed SDGs across the 3 main sectors society, economy and environment. However, this number is improved when you narrow the scope to environment only at 93% [1].Building on this broader context, this study focuses specifically on analyzing average temperature trends and their projections. Using polynomial regression and a correlation matrix, the analysis explores the temporal dynamics of temperature changes and their relationships with environmental factors, revealing both expected and some counterintuitive patterns.

III. Methodology

The datasets used in this study was sourced from Kaggle, a platform providing diverse, open-access datasets. The first includes annual temperature change metrics for various countries, spanning from the 1960s to recent years. Key features of the dataset include unique identifiers for each country and year, along with corresponding temperature change values. These features allow for a temporal and geographical analysis of climate trends, forming the foundation for the regression modeling performed in this study. This was done by means of a python notebook utilizing several libraries such as pandas and NumPy for data processing, seaborn/matplotlib for visualization and scikit-learn for machine learning tasks

Prior to analysis, the datasets were preprocessed to handle missing values and make the set readable. While imputation is a common method of handling missing data, it was decided against to not misrepresent a countries contribution to the data. Instead deciding to leave out entries with missing data from the analysis by simply dropping them from the set before implementation. Other changes to the data include revamping naming conventions and dropping unnecessary columns such as redundant identifications of countries and time frames. Narrowing the data to entries of temperature change, analysis began with graphing the probability density function of temperature increases over the entire dataset. Then refining the function over the last 20 recorded years from the set showed the trend in data. To better visualize these trends, there were boxplots plotted of each year's average over an average line trending toward a reference line suggested to be a critical point in global warming. The critical point in

question is an average global increase of 1.5 degrees Celsius from that of the pre-industrial period. However, this set only goes back to 1961 and only so far as 2019 providing a representative, yet incomplete analysis of our progression toward this point.

The analytics of the current data led me to consider the future of global warming. Linear regression was used to predict values of temperature change to be assessed based on multiple accuracy metrics. The regression predicted values were compared with the actual values and the differences were visualized to provide more context to the accuracy of the model. However, polynomial regression was used to perform the predictions on future values through the year 2050. Polynomial regression is better suited to capture nonlinear relationships between independent and dependent variable such as temperature change over decades. In this case quadratic regression was intentionally selected to find a balance in model complexity and interpretability. Higher degree regression can overfit the data, while quadratic aptly captures higher degree relationships such as an acceleration of temperature change over time.

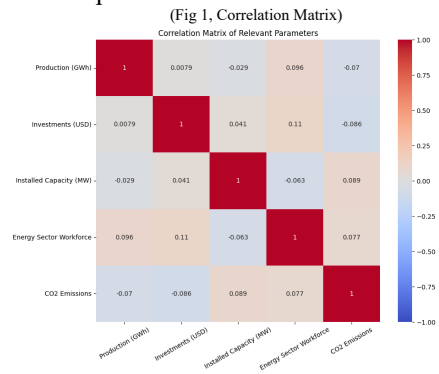
Further analysis on green energy data and its socio-environmental interactions was done. Fortunately, this dataset did not contain any missing values and only required basic preprocessing like sorting and unit conversion for readability. Narrowing down to the U.S. after reviewing investments by nation. Visualizing the differences between how much green energy production covers our overall energy consumption. Then a heat mapped correlation matrix was utilized to investigate the relationships between several features within the set. To assess these correlations accurately, logarithmic normalization was used on the monetary values from the data and then standard scalar was implemented across all data before generating the matrix.

IV. RESULTS

The initial probability density function provided insight into the trends of temperature changes over the recorded time frame of the dataset. Analyzing average change from 1961-2019 visualized and calculated a righthanded skewness of 0.12 which is modest, however it prompted my usage of the same function over the final 20 years of the data. The second PDF provide a skewness of 0.54 showing an increase of 0.42 which demonstrates a clear acceleration in temperature change. The plotting of these averages shows a gradual increase via boxplots of yearly averages reporting a final increase in temperature of 1.44C nearly reaching the posited critical temperature change of 1.5C increase from the pre-industrial period. Future predictions were needed to ascertain when we are predicted to reach this mark. Utilizing linear regression provided a root mean squared error score of 1.0095, meaning that each prediction is roughly 1.01 units away from the actual values. While this accuracy is fine, it was far enough off to research other methods of achieving more accurate predictions. Consequently, polynomial regression was applied in fundamentally the same way after transforming the independent variable as a quadratic feature. Plotting these predicted averages as a continuation of the previous graph of average temperature increases visualized the persistence of an accelerated trend. This model indicated we had surpassed the critical point in 2022 and produced a final increase of 3.15C by 2050.

To analyze green energy production metrics, 10 countries were ranked by investment to see which country might provide a specific interest in renewables. While the U.S. being only the 4th highest was a surprise, a decision was made to evaluate its data due to relevance. As such, overall energy consumption and green energy production totals by year were plotted, calculating an average coverage of 31.39% our consumption.

(Fig 1, Correlation Matrix)



Finally, a heat mapped correlation matrix (Fig1) was produced measuring the relations of several socio-economic features. The features analyzed were, production, investments, installed capacity, green energy jobs and CO2 emissions providing a perplexing array of correlation values. In an attempt to achieve more expected correlations, logarithmic normalization and standard scalars were applied to relevant features. Generating values closer to expectation, although weaker than expected. Some highlighted values were the connections between investment and CO2 emissions at -0.086 and between energy production and green energy jobs at 0.096. The first meaning that as investment increases, CO2 emissions go down and the second supporting that as production goes up, jobs also rise.

V. Conclusions

The results of this study exhibit both its value as a way to measure and predict sustainable development goals, as well as highlight a need for future development in the integration of AI and the environment. This can be seen in the comprehensive analysis of temperature change and its future predictions and the counterintuitive or weak correlation responses for relevant socio-economic features.

Some limitations faced in this study arose due to incomplete or inaccurate representation of values in the second dataset which is suspected as the reason for the misleading correlation values. Additionally, attempts were made to further analyze the optimization of renewable energy production using a nations natural resources via methods that will be expanded on in my future works but were left out of my analysis due to their inaccuracies. My own knowledge of natural sciences limited my ability to properly set up models for features like solar irradiance and energy potentials.

VI. Future works

As mentioned, attempts at further analysis of renewable energy production were made that were not complete enough for this study. Producing a K means map to view countries based on their predominate relevant resources was one such omission. This was in an attempt to sort countries for optimization algorithms using data such as installed capacity, financial availability for improvement, solar irradiance and weather data. Other optimization models could be possible if more in depth data on these methods is sourced and properly handled. Further research into the way these renewable energy sources are engineered could aid in the ability to model viable strategies for increased capacity Another potential work of relevance would be to focus on the features of this data that actually drive a nation to change such as their financial impact. Deriving a method of showing renewable energy optimization as a metric of money to be made by a nation could spark real change in policy and investment for the future.

VII. References

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